

Causal Inference: Panel Data and Fixed Effects Models

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So far

- ▶ What happens if we cannot randomize?
- ▶ Selection on observables and the prospects for causal inference
- ▶ Matching
- ▶ Regression

Let's begin with a recap!

Recap

Strategies for estimating effects of treatments so far:

- ▶ Randomize treatment and take the DIGM
- ▶ Identify and control for confounding variables such that the CIA (Conditional Independence Assumption) holds

Today: Use observations at more than one point in time

Types of samples

- ▶ Cross section: observe many units at one point in time. **So far!**
- ▶ Time series: observe one unit at many points in time
- ▶ Repeated cross sections: observe many units at many points in time, and the units are (potentially) different over time
- ▶ Panel data: observe many units at many points in time, and the units are the same over time

Repeated Cross Sections

- ▶ At one point in time ($t = 1$) we take a random sample of n_1 individuals ($i = 1 \dots n_1$)
- ▶ At another later point in time ($t = 2$) we take another random sample of **potentially different** n_2 individuals ($i = 1 \dots n_2$)
- ▶ Every period we repeat the process of obtaining a cross section
- ▶ This data is not panel data because the individuals are not the same over time (or they are and we still do not treat them as such)
- ▶ Why would we want to have cross sections at different points in time? Larger N, study whether the slope estimate changes over time as a result of policy change (or shock in time)
- ▶ Often the case in **surveys**

Panel Data

- ▶ At one point in time ($t = 1$) we take a random sample of n individuals/units ($i = 1...n$)
- ▶ At another later point in time ($t = 2$) we sample again the same n individuals/units ($i = 1...n$)
- ▶ Every period we repeat the process of **sampling the same individuals/units**: the same individuals/units are measured at different points in time
- ▶ If every individual appearing in $t = 1$ also appears in $t = 2, 3, \dots$: **balanced panel**

Country ID	Year	Var1	Var 2
Afghanistan	2018	x1	y1
Afghanistan	2019	x1	y1
Afghanistan	2020	x1	y1
Croatia	2018	x2	y2
Croatia	2019	x2	y2
Croatia	2020	x2	y2
United Kingdom	2018	x3	y3
United Kingdom	2019	x3	y3
United Kingdom	2020	x3	y3

Empirical example: Number of Women MPs and Domestic Abuse Legislation

Let's imagine we are interested in the question whether the number of **Women MPs** change the nature of legislation, particularly whether it changes Domestic Abuse legislation passed. Suppose we observe 20 countries over a 20 year period.

$$\text{Domestic Abuse Legislation}_{i,t} = \alpha + \beta \text{Number of Women MPs}_{i,t} + u_{i,t}$$

- ▶ What are the potential omitted variables?
- ▶ What about time?
- ▶ Has the number of women MPs changed through time? Has domestic abuse legislation changed through time?

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Empirical example: Number of Women MPs and Domestic Abuse Legislation

Let's imagine we are interested in the question whether the number of **Women MPs** change the nature of legislation, particularly whether it changes Domestic Abuse legislation passed? We observe 20 countries over a 20 year period (2000-2020).

$$\text{Domestic Abuse Legislation}_{i,t} = \alpha_1 + \alpha_2 \text{decade2010}_t + \beta \text{Number of Women MPs}_{i,t} + u_{i,t}(1)$$

- ▶ What is the interpretation of $\hat{\alpha}_1$?
- ▶ What is the interpretation of $\hat{\alpha}_2$?
- ▶ Does the inclusion of decade fixed effects help with selection bias concerns?

Empirical example: estimation

Impose Parallel Slopes

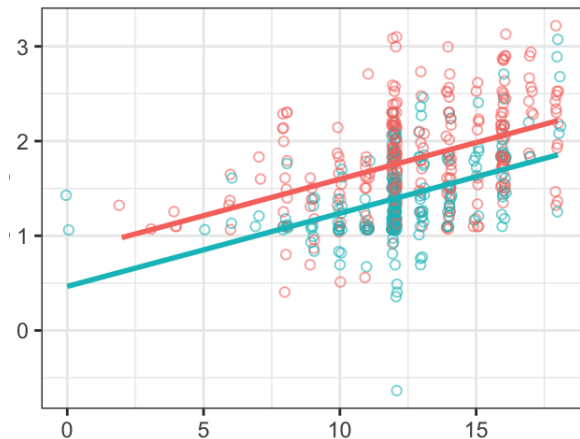


Figure: Different intercepts for two decades, same slope

Empirical example: Number of Women MPs and Domestic Abuse Legislation

We could also add individual year fixed effects for every year in our sample:

$$\text{Domestic Abuse Legislation}_{i,t} = \alpha_1 + \alpha_2 y_{2002,t} + \alpha_3 y_{2003,t} + \alpha_4 y_{2004,t} + \alpha_5 y_{2005,t} + \alpha_6 y_{2006,t} + (\dots) + \alpha_{20} y_{2020,t} + \beta \text{Number of Women MPs}_{i,t} + u_{i,t} \quad (2)$$

This is equivalent to writing:

$$\text{Domestic Abuse Legislation}_{i,t} = \alpha_1 + \delta_t + \beta \text{Number of Women MPs}_{i,t} + u_{i,t} \quad (3)$$

- ▶ What do the δ s estimate?
- ▶ Does the inclusion of year fixed effects help with selection bias concerns?
- ▶ Control for any general time varying unobservable factors. The variation used to estimate the effect is the number of women MPs in a **given year** but **across countries/units**.

Panel Data: First Difference Estimator

- ▶ Panel data can help control for some omitted variable bias
- ▶ We sample the same individuals over two periods and assume that:
 - ▶ the intercepts are different for the two periods
 - ▶ the slopes (causal effects of interest) are the same over the two periods
- ▶ We can write a model for the **first** period:

$$y_{i,1} = \beta_{0,1} + \beta_1 x_{i11} + \beta_2 x_{i12} + (\dots) + \beta_k x_{i1k} + \epsilon_{i,1}$$

- ▶ We can write a model for the **second** period:

$$y_{i,2} = \beta_{0,2} + \beta_1 x_{i21} + \beta_2 x_{i22} + (\dots) + \beta_k x_{i2k} + \epsilon_{i,2}$$

Panel Data: First Difference Estimator

Assume that the error term in period t , consists of two parts:

$$\epsilon_{i,t} = a_i + v_{i,t}$$

where a_i : part that is constant over time (fixed effect)

$v_{i,t}$: part that changes every period (idiosyncratic error)

Examples:

For individuals:

- ▶ a_i : genetic material, ability, character,... **time invariant**
- ▶ $v_{i,t}$: luck, experience, performance...

For countries:

- ▶ a_i : geography (coastal access), colonial history, ... **time invariant**
- ▶ $v_{i,t}$: economic or political shocks...

Panel Data: First Difference Estimator

The **first** period:

$$y_{i,1} = \beta_{0,1} + \beta_1 x_{i11} + \beta_2 x_{i12} + (\dots) + \beta_k x_{i1k} + \underbrace{a_i + v_{i1}}_{\epsilon_{i,1}}$$

The **second** period:

$$y_{i,2} = \beta_{0,2} + \beta_1 x_{i21} + \beta_2 x_{i22} + (\dots) + \beta_k x_{i2k} + \underbrace{a_i + v_{i2}}_{\epsilon_{i,2}}$$

Subtracting one period from the other:

$$\Delta y_i = \delta + \beta_1 \Delta x_{i1} + \beta_2 \Delta x_{i2} + (\dots) + \beta_k \Delta x_{ik} + \Delta v_i$$

The fixed effect a_i has disappeared. Anything observable or not observable that was a confounder and is time invariant at the unit level is not a concern anymore.

Fixed Effects Estimator for Panel Data

Assume that we have a panel dataset over twenty years.

→ **Fixed Effects Estimator:**

- ▶ Regression with Entity/Unit Fixed Effects: Method for controlling for omitted variables in panel data when omitted variables vary across individuals but not over time.
- ▶ This is similar to a first-differences estimator (equivalent if $t=2$).
- ▶ Regression with Time Fixed Effects: Method for controlling for omitted variables in panel data when omitted variables vary across time but not across individuals.
- ▶ **Examples?**

Regression with Time Fixed Effects

- ▶ Time fixed effects can control for variables that are constant across entities but evolve over time.
- ▶ For example, in a panel of UK electoral districts/regions this may be variables like UK annual GDP, financial crisis, Brexit, etc...
- ▶ If an omitted variable X_t is unobserved, its influence can be exogenised because it varies over time but not across units.
- ▶ This means specifying a model that has a different intercept λ_t for each time period (T-1 binary indicators). Then the terms $\lambda_1 \dots \lambda_T$ are known as time fixed effects.

$$y_{i,t} = \beta_0 + \beta_1 X_{i,t} + \delta_2 Q2_t + (\dots) + \delta_T Q_T + u_{i,t}$$

- ▶ Where $\delta_2, \dots, \delta_T$ are unknown coefficients and where $Q2_t = 1$ if $t = 2$, and so forth.

Regression with Entity Fixed Effects

- ▶ Entity fixed effects can control for variables that are constant across time but vary across entities.
- ▶ That is, *unit-specific, time-invariant* heterogeneity is not a problem anymore.
- ▶ This means specifying a model with entity specific intercepts $\alpha_1, \alpha_2, \dots, \alpha_n$

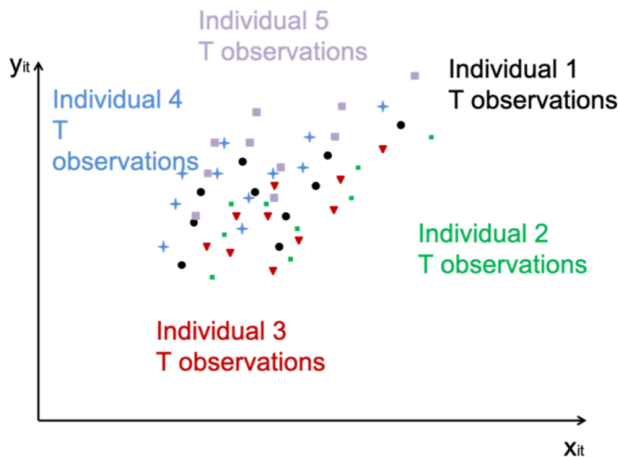
$$y_{i,t} = \beta_0 + \beta_1 X_{i,t} + \gamma_2 D2_i + (\dots) + \gamma_n Dn_i + u_{i,t}$$

- ▶ where $D2_i = 1$ if $i = 2$, $D3_i = 1$ if $i = 3$, etc..
- ▶ Equivalent to writing the below where $\alpha_1 \dots \alpha_n$ are entity specific intercepts.

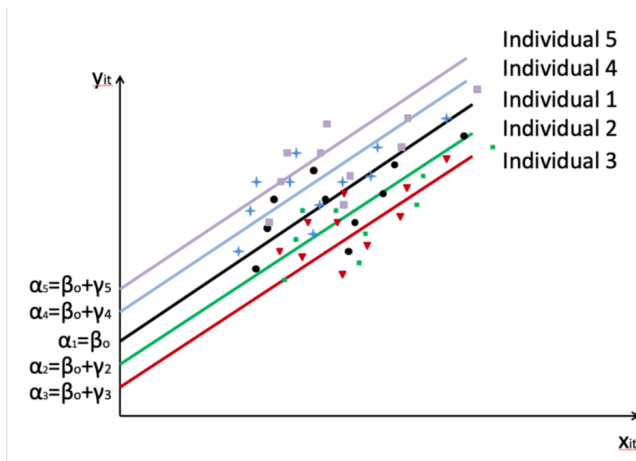
$$y_{i,t} = \beta_1 X_{i,t} + \alpha_i + u_{i,t}$$

Regression with Entity Fixed Effects: Example of Data

We have information for all individuals over time:



Regression with Entity Fixed Effects: Example of Estimation



Regression with Entity Fixed Effects and Time Fixed Effects

Imagine we go back to our original example of examining (a similar) question of the effect of a women policymakers on the representation of women's issues across a panel of US states and 20 years:

$$\text{Domestic Abuse Funding}_{i,t} = \alpha_i + \delta_t + \beta \text{Congresswoman}_{i,t} + u_{i,t} \quad (4)$$

- ▶ What are we controlling for with α_i and δ_t ?
- ▶ Give examples of selection bias that has been resolved.
- ▶ What is the leftover variation in $\text{Congresswoman}_{i,t}$ used to estimate β ?
- ▶ What could be the sources of remaning selection bias?

Regression with Entity Fixed Effects and Time Fixed Effects and Potential Outcomes Framework

We need to satisfy: $P(D_i = 1) \perp\!\!\!\perp Y_0, Y_1$

Experiments

- ▶ Randomization ensures unconfoundedness without selection on observables: $P(D_i = 1) \perp\!\!\!\perp Y_0, Y_1$ which translates into:
- ▶ $E(Y_0|D = 1) = E(Y_0|D = 0)$

Panel data fixed effects models (observational studies)

- ▶ Unconfoundedness can be assumed to hold only after conditioning on a set of pre-treatment variables and time and entity fixed effects, which translates into:
- ▶ $E(Y_0|D = 1, X_{i,t}, \alpha_i, \delta_t) = E(Y_0|D = 0, X_{i,t}, \alpha_i, \delta_t)$

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Diagnostic Tests and SEs

Assumptions for standard errors

What does the standard error mean?

$$se(\hat{\beta}) = \sqrt{\frac{\hat{var}(e_i)}{n \times \hat{var}(X_i)}}$$

Basic assumptions behind OLS standard errors:

- ▶ Variance of regression errors independent of X (homoskedastic)
- ▶ Regression errors independent of each other (uncorrelated across observations)

Is second assumption likely to be met in the panel data case?

Problem with repeated observations: Addressing correlations among errors

- ▶ We cannot make the assumption anymore that regression errors are independent of each other - likely they are correlated within time or within unit.
- ▶ We then make the common assumption in panel data that regression errors are independent **except within clusters** → **cluster-robust standard errors**.
- ▶ Clustered standard errors allow for heteroskedasticity and autocorrelation within an entity, but treat the errors as uncorrelated across entities.
- ▶ Typically, the clusters are the panel units (e.g. municipalities, states). It doesn't work with 2 clusters → usually need at least thirty or more clusters.

See `estimatr` or `lfe` packages (in R).

In Stata, see `cluster()`.

Random Effects Models

We begin with a similar unobserved effects model as before:

$$y_{i,t} = \beta_0 + \beta_1 X_{i,t,1} + \beta_2 X_{i,t,2} + \alpha_i + u_{i,t}$$

- ▶ We explicitly include an intercept so we can make the assumption that the unobserved effect α_i has zero mean.
- ▶ Suppose we think that α_i is uncorrelated with each explanatory variable in all time periods.
- ▶ $cov(x_{i,t,j}, \alpha_i) = 0$ for $t = 1..T; j = 1...k$.
- ▶ In this case, β_j is unbiased and can be consistently estimated by using a single cross section OLS ($E[u|X] = 0$ is satisfied). But eliminating α_i from the specification will result in inefficient estimators.

Diagnostic Tests: F-Test

Are the fixed effects predictive of the outcome?

$$y_{i,t} = \beta_1 X_{i,t} + \alpha_i + u_{i,t}$$

- ▶ Specify H_0 and H_1 .
- ▶ $H_0 : \alpha_1 = \alpha_2 = \alpha_3 = \dots = \alpha_n = 0$
- ▶ $H_1 : \alpha_1 \neq \alpha_2 \neq \alpha_3 \neq \dots \alpha_n \neq 0$ (or at least one)
- ▶ Run the restricted model and retrieve SSR: $y_{i,t} = \beta_1 X_{i,t} + u_{i,t}$
- ▶ Conduct F-test with a set F critical value. Reject the null hypothesis if $F > F_c$

$$F = \frac{(SSR_R - SSR_{UR})/q}{SSR_{UR}/n - k - 1}$$

Diagnostic tests: Hausman Test

- ▶ The Hausman test can be used to differentiate between fixed effects model and random effects model in panel analysis. In this case, Random effects (RE) is preferred under the null hypothesis due to higher efficiency, while under the alternative Fixed effects (FE) is unbiased and at least as consistent and thus preferred.
- ▶ You can run a Hausman test (which tests whether the unobserved individual effects are correlated with the regressors, the null is they are not). If the p-value indicates statistical significance, then you choose fixed effects (since the unobserved individual effects are correlated with the regressors).
- ▶ $H_0 : cov(X_{i,t}, \alpha_i) = 0$
- ▶ $H_1 : cov(X_{i,t}, \alpha_i) \neq 0$